

Depth-Weighted Multi-View Fusion for Mesh Vertex Features

- Depth point (~57k)
- Mesh vertex (~7k)

Multi-View Projection

V
Mesh Vertex
(world space)

View 1

z_proj=3.5
z_map=3.4

w=0.65

View 7

z_proj=3.8
z_map=3.7

w=0.67

View 14

z_proj=4.2
z_map=2.1

w=0.00

View 20

z_proj=3.6
z_map=3.0

w=0.08

$$w_{\text{depth}}(v) = \exp\left(-\alpha_d \left|1 - \frac{z_{\text{map}}}{z_{\text{proj}}}\right|\right)$$
$$\alpha_d = 15.0$$

Weighted Aggregation

f_1

f_2

f_3

f_4

$$\frac{\sum w \cdot f}{\sum w}$$

f(v)

Final 128-d
Feature Vector

- Depth Consistency
- Visible (w ≈ 1.0)
 - Partial (w ≈ 0.5)
 - Occluded (w ≈ 0.0)